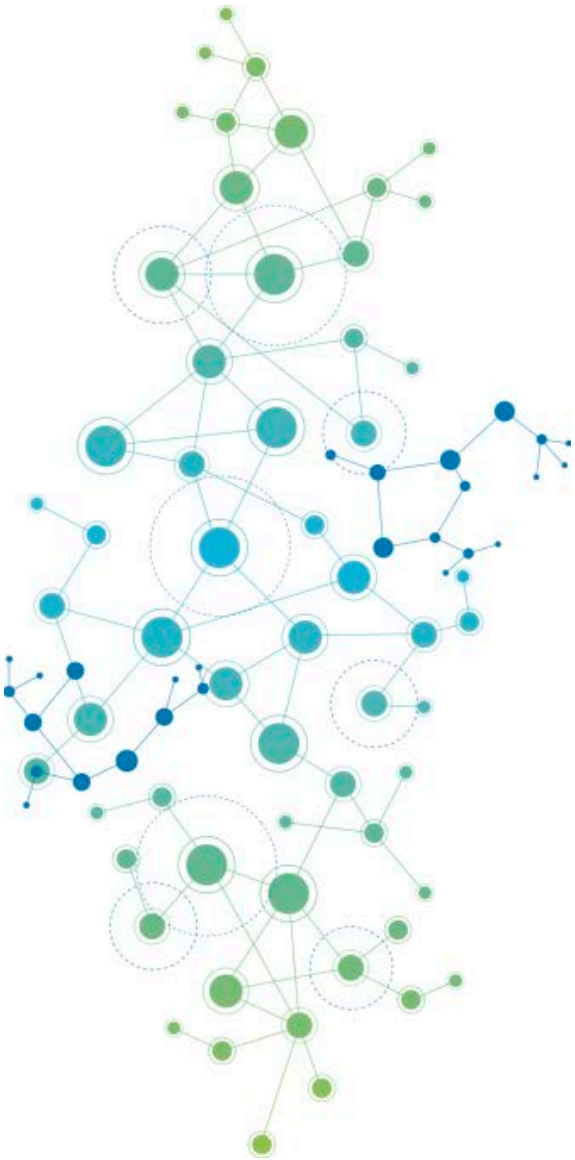

Communism vs. Capitalism

QUANTITATIVE ANALYSIS AT THE NEW SCHOOL

Hart Traveller | Statistics with SPSS

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INTRODUCTION

Abstract

The goal of this study is to compare a series of dependent variables to the preferred economic system of a sample of students at The New School. We will create a hypothesis for each of the dependent variables about the level of correlation between the dependent and independent variables, and then analyze the data to confirm or reject our hypotheses. In other words, we're trying to figure out if some of the differences between capitalist and communist students at The New School can be quantified.

Study Overview

Why did I choose my topic?

Over the past year I have met many communists at The New School, and I figured that for this study using the large communist population would be a good resource to see if it was possible to quantify the differences between communists and capitalists.

Why is my topic important to others?

This topic specifically (the difference between communists and capitalists) is important because by human nature (historically speaking), we have a knack for categorizing groups of people, and then attacking the categories that we see as "the other". "Communist" and "capitalist" are two major categories we have invented that are often part of political discourse on a local and global scale. I think quantifying some of the differences and exploring how we are potentially less different than we pretend to be would be of significance to many people. This is important to others because this is what some groups of people are actively trying to do, and what some groups are actively trying to prevent, and no matter which camp you fall into, a study like this might be interesting to read.

How was the data gathered?

I created a survey with 8 questions and sent it out to several other classes and several of my friends. Originally, I was going to collect the survey data as a convenience sample, however I was able to access The New School student directory through my New School account and from there I sent the survey out to several thousand students. This part of the survey (from which I collected most of the data) was done using random sampling, in that only people whose names start with "A" had the survey sent to them, that is to say, everybody whose name starts with an "A" at The New School received a copy of the survey.

I was not supposed to do this though, and I ended up having to attend a disciplinary meeting. I got off with just a warning though (and a much larger sample size).

What was the explanatory variable?

The explanatory variable was the respondents preferred economic system.

What was the response variable?

I examined a series of different response variables. These included:

- Gender
- Age
- Risk tolerance
- USA sentiment
- Belief in hard work for success
- Parental strictness
- Values
- Preference for equality or freedom

Why did you choose these response variables?

<u>Gender</u>

This was a required piece of data to collect for the project.

<u>Age</u>

This was a required piece of data to collect for the project.

<u>Risk tolerance</u>

Theoretically, by nature a capitalist society demands more risk, as individuals must take risks to improve their own lives. In communist society, a lower degree of risk is (theoretically) needed to live, because the state handles your affairs and livelihood. This is an oversimplification, but is it possible there are differences in risk tolerance levels between communists and capitalists?

<u>USA sentiment</u>

There is a stereotype that capitalists approve of the United States and its effect on the world more than communists. This makes sense for the simple fact that the US is a capitalist country, but can it be quantitively proven?

<u>Belief that working hard equates to success</u>

Theoretically, in a capitalist meritocracy such as the US, the harder you work, the more success you will see in your life. On the other hand, in a communist society such as the Soviet Union the harder you work, the same amount of daily bread rations you get. Communists that I talk to often speak of how corruption is rampant in the US meritocracy. If you are on the bottom you are stuck there, while if you are on the top you are guaranteed to succeed no matter what. Meanwhile I occasionally run into dedicated patriots that believe intensely in the capitalist meritocracy and think that those who do not believe are just lazy and not willing to put in the work it takes to succeed. Can the difference in opinion between these two groups be quantitively measured?

Parental strictness

Communism is often associated with big government, all-encompassing, and invasive authority. Capitalism is often associated with small government and little authority. The first authority most us deal with in our life is our parents. Is it possible that there is a correlation between the economic system people prefer and their parents level of strictness?

Values

The values question is meant to measure what values the respondent holds. There is a selection of values from which the respondent is allowed to choose three. The choices are as follows:

- Family
- Romance
- Money
- Power
- Fame
- Beauty
- Peace
- Meaning
- Fulfillment
- Other…

The goal of this question is to measure the statistical correlation between the respondents preferred economic system and their values.

Equality vs. Freedom

While this question is similar to the previous question I nonetheless wanted to separate it. I wanted to ask about these two specific values because of a theory posited in a book I was recently reading. In the book, the author posits that during the cold war and in politics today, the biggest warring values were not communism and capitalism, nor today the right and the left, but instead whether people hold the value of universal equality or the value of individual freedom in higher esteem. I wanted to test this and see if there was a correlation between these specific values and economic system preference.

DATA ANALYSIS

Confidence in distribution of opinions

To calculate the accuracy of our results (our results being the distribution of people that believe in capitalism/communism), we need to perform a confidence interval calculation using the confidence interval formula. First, we need to gather the required figures from the data using SPSS. These are the mean, standard deviation, and sample size. I have shown the results below. Note that I've calculated the mean and standard deviation by assigning dummy variable values to economic preference, with 1 being capitalism and 0 being communism.

	Total
Mean	0.6827
Standard Deviation	0.4665
Sample Size	105

Now we need to plug our data into the confidence interval formula.

X +/- [(Z*s)/ √n]

0.6827 +/- [(1.96*0.4665)/ √105]

0.6827 +/- [0.91434/10.247]

0.6827 +/- 0.0892

Therefore our 95% confidence interval is (0.5935, 0.7719), or in other words, we can be 95% confident that the mean of the dataset lies between a lower limit of 0.5935 and an upper limit of 0.7719. To phrase this another way: we are 95% confident that 59% to 77% of students attending The New School prefer capitalism over communism.

Gender

Null Hypothesis: There is no correlation between gender and economic system preference.

Alternative Hypothesis: There is a correlation between gender and economic system preference.

(Side note: While most respondents identified as either male or female, one respondent identified as "other". They have been excluded from the data analysis of gender, as it would be unfair to make a broad statement about the economic system of preference of non-binary people based on a sample size of 1.)

To conduct hypothesis testing, we first need to calculate the mean, standard deviation, and sample size of the economic system preference for males, and for females in our sample, the resulting values of the calculation are in the table below.

	Male	**Female**	**Total**
Mean	0.6286	0.7143	0.6827
Standard Deviation	0.4902	0.4550	0.4665
Sample Size	35	70	105

Given that the data we have in this case is binary, either male or female, the mean and standard deviation in this case were calculated using dummy variables, where capitalist was 1 and communist was 0. The mean then refers to the percentage of respondents (either male or female) that identified as capitalists. For instance, approx. 63% of males identified as capitalists, and approx. 71% of females identified as capitalists. Of the total sample, approx. 67% of respondents identified as capitalist. The standard deviation refers to the average distance from the mean. Note as well that for this test, we are setting the significance level at 0.05.

If the null hypothesis is true and there is no correlation between gender and economic system preference, then we would expect to see no difference between the rates of communism and capitalism when comparing males and females, ie: the mean should be approx. the same for males and females. This is not true in our sample (of students at The New School), so it is possible there is a correlation between these two variables, but to properly ascertain that, we need to calculate the probability the null hypothesis is true (and by default, the probability that it's false as well).

So, if we assume that the null hypothesis is true, what is the probability that we would've gotten the results we did in our sample? If the probability is really small, then the null hypothesis is probably not true and thereby we can reject it. To get as close to the population mean as possible, we'll be using the mean of the total as a hypothetical population mean. We'll use males to test against the population, because even though the sample size is smaller, due to the smaller number of males they effect the total population mean much less.

Ho (Null Hypothesis)	No Correlation	Mean = 0.6714
H1 (Alternative Hypothesis)	Correlation	Mean != 0.6714

First, we will assume that the null hypothesis is true. If it was, what is the probability that we would have gotten the result we did?

If our null hypothesis is true, then the mean of our sampling distribution would be equal to the mean of our population distribution which would be 0.6714. What is the standard deviation of our sampling distribution?

$STDEV_{\text{sampling distribution}} = STDEV_{\text{population distribution}}/\sqrt{\text{Sample size}}$

$STDEV_{\text{sampling distribution}} = 0.4902/\sqrt{35}$

$STDEV_{\text{sampling distribution}} = 0.0829$

Now we need to figure out how many standard deviations away from the population mean our sample mean was.

$Z = (\text{Sample mean} - \text{Population mean})/ STDEV_{\text{sampling distribution}}$

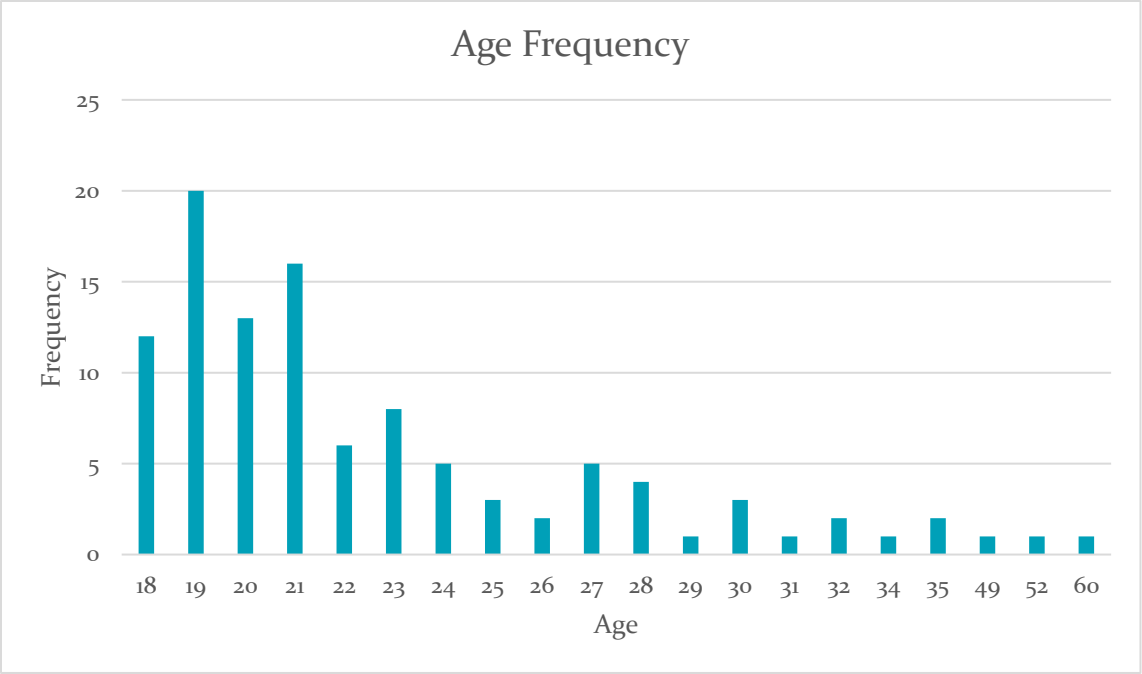
$Z = (0.6286 - 0.6827)/0.0829$

$Z = -0.65$

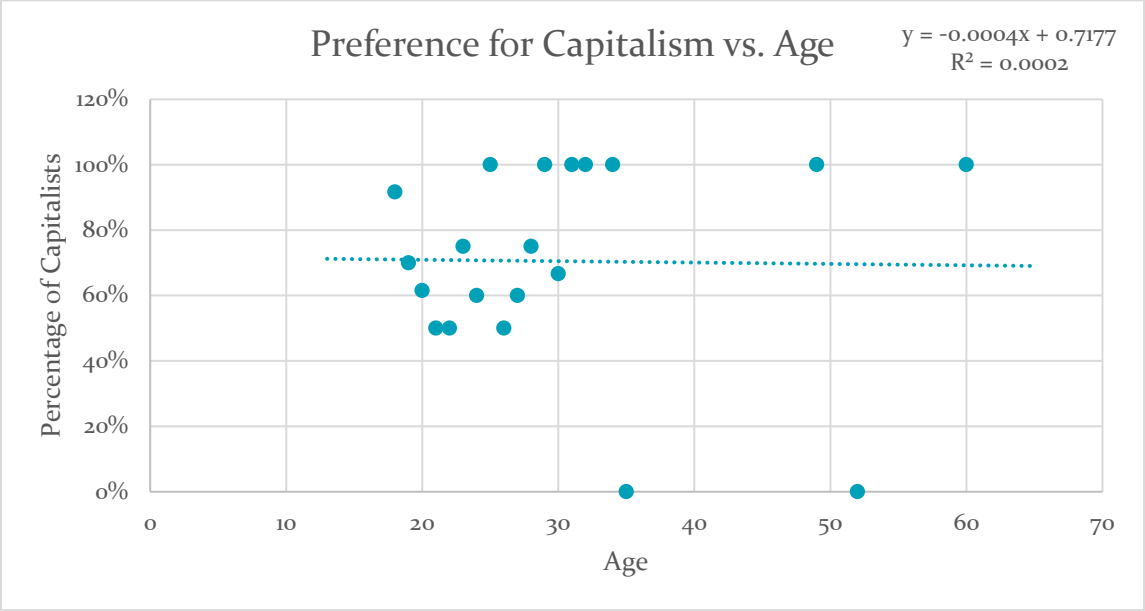
Now that we've found our z score, know that our sample mean is -0.65 standard deviations away from our population mean. If we look at a normally distributed z table, we find that the area to the left of our sample is 0.2578. To reject our null hypothesis, we would need a value of 0.05 (our original significance value) or less. Therefore, we cannot reject the null hypothesis.

Age

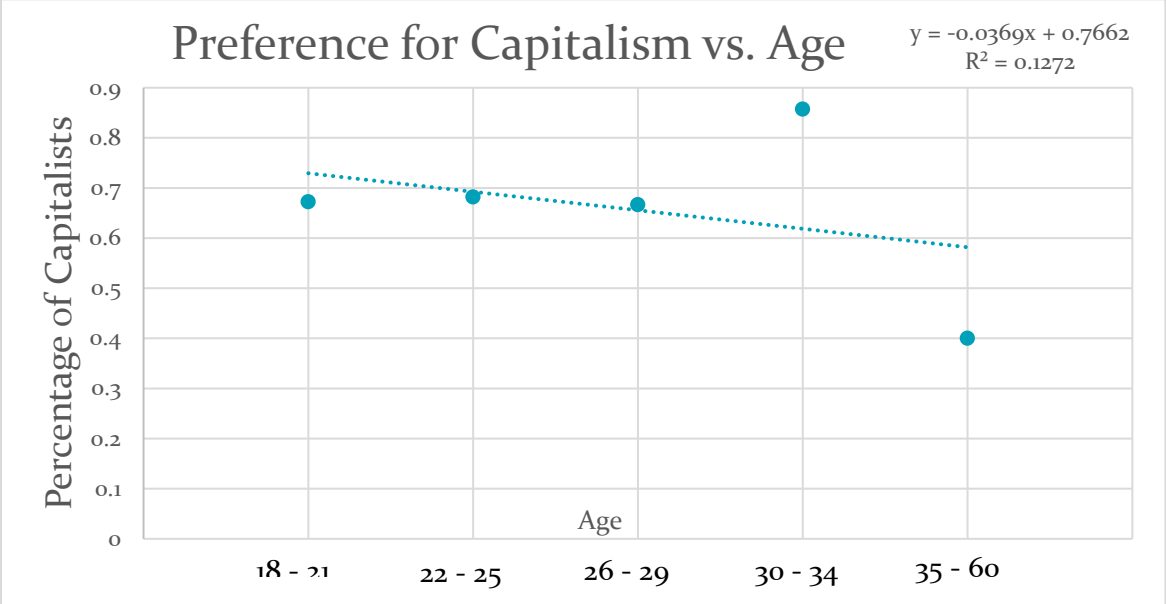
Age was significantly more difficult to draw out meaningful conclusions from due to the inconsistencies in the different age brackets available.



As we can see in this frequency chart, there was a much higher percentage of younger respondents, with older respondents being rare. As such the data is heavily biased towards the younger respondents. Even when reweighting the data to account for differences in age frequency, problems arise. When finding correlation between economic system and age, certain older ages appear to entirely agree or disagree with an economic system, which is obviously not true but simply due to the lack of data at certain ages. With that said, I still found it interesting to perform a regression analysis on it and I have included the results nonetheless.



After performing a regression analysis on the individual ages, we find just about the smallest amount of correlation possible. The R^2 value, which we might expect to be 0.8 for a strong correlation and 0.5 for a weak correlation, in this case is 0.0002. Even though we had a limited amount of data in this case, there certainly seems to be a trend towards absolutely no discernable correlation.



If we increase the age brackets so the data points are they are the average of every 4 consecutive ages we start to see a potential correlation emerge, although it is worth noting this data is skewed due to the last age group, which contains a significantly bigger age bracket. Nonetheless, the correlation is still incredibly weak, and we can say with great confidence that there is little to no correlation between age and economic system preference.

Risk Tolerance

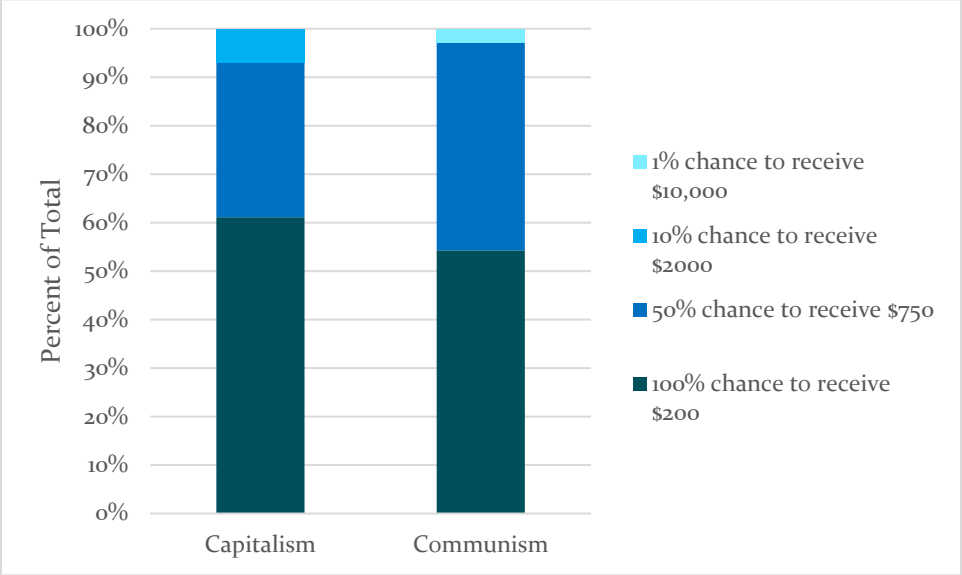
The following frequency table shows the risk tolerance levels for those who identified with capitalism as their preferred economic system.

Risk Tolerance	Frequency
No risk tolerance	54%
Low risk tolerance	43%
Medium risk tolerance	0%
High risk Tolerance	3%

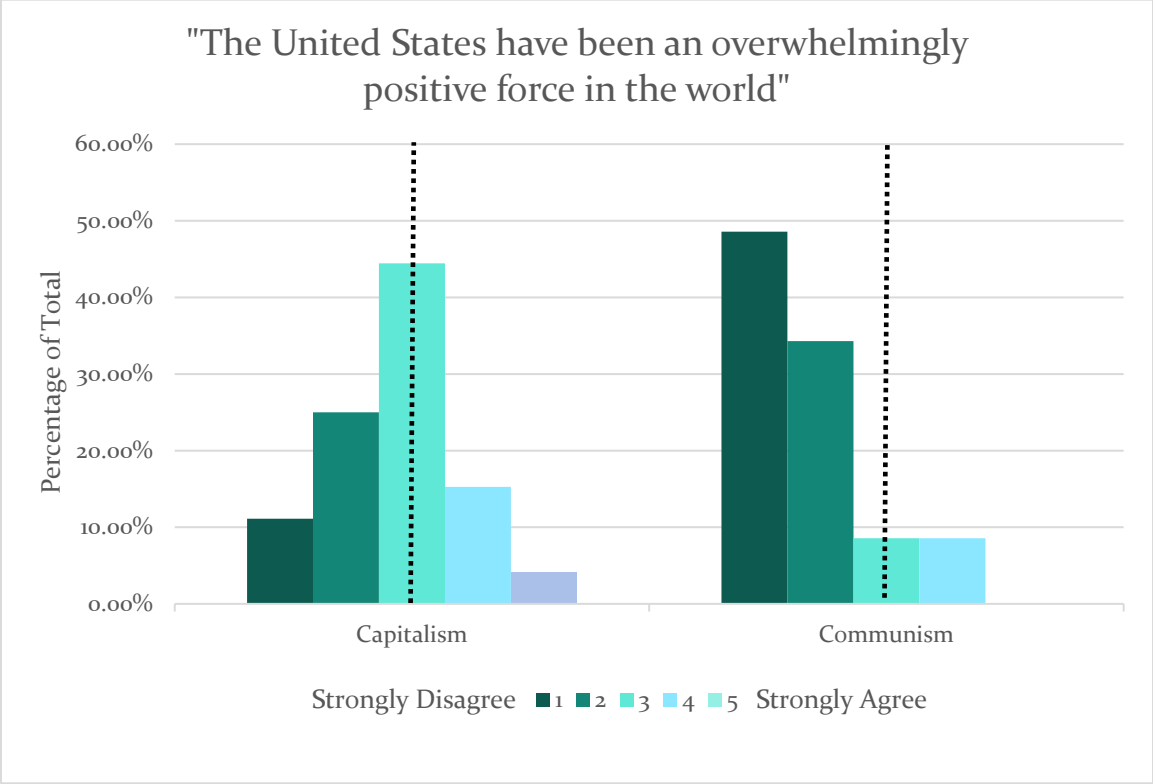
The following frequency table shows the risk tolerance levels for those who identified with communism as their preferred economic system.

Risk Tolerance	Frequency
No risk tolerance	61%
Low risk tolerance	32%
Medium risk tolerance	7%
High risk tolerance	0%

There is a negligible difference in risk tolerance between those who identify with capitalism as their preferred economic system compared to those who prefer communism. We can see this more clearly in the following stacked column chart.



Is the USA an [overwhelmingly] positive global force?



The above chart is a histogram that shows the distribution of opinions for capitalists and the distribution of opinions for communists. The black dotted line marks the halfway mark in the data and is meant to help visualize the skewness in the data.

Looking at the chart, it's clear while capitalist respondents are close to being normally distributed, communist respondents are quite positively skewed. It is easy to see that significantly more communists disagree with this statement than capitalists. Let's see if we can directly quantify this using the descriptive statistics available to us.

The descriptive statistics differ quite a bit between the two samples here. I've included them below.

Capitalist Stats	*Stats*	*Communist Stats*	*Stats*
Minimum	1	Minimum	1
Q1	2	Q1	1
Median	3	Median	2
Q3	3	Q3	2
Maximum	5	Maximum	4
Mean	2.76	Mean	1.77
Mode	3	Mode	1
Stand.Dev	0.99	Stand.Dev	0.94
Kurtosis	-0.14	Kurtosis	0.59
Skewness	0.04	Skewness	1.16

Looking at the 5-number summary, the communist distribution leans much more heavily towards the lower numbers. The respective means of the two samples show this clearly, with a difference of a whole point between the two. The modes of the two samples show that while a capitalist is most likely to be ambivalent about the statement, a communist is most likely to strongly disagree. While both samples are platykurtic, the communist sample is closer to resembling a mesokurtic normal distribution than the capitalist sample. The reason for this is due to the higher clustering of communist sentiment around the lower end of the scale, while capitalists are more likely to be varied in their opinions (at least in this sample). As a communist in our sample, you are more likely to be decisive about your answer, while capitalist sentiment is more likely to be spread across the entire distribution. The skewness reveals the greatest difference however. While the capitalist sentiment is only slightly positively skewed, the communist sentiment is heavily positively skewed, so much so that the overwhelming majority of the sample disagrees with the statement. We can get a better view of this by examining a frequency chart. (Another interesting note, our maximum for the communist sample was 4, which means not a single communist chose "strongly agree" for the statement.)

Capitalist Agreement	Frequency	Communist Agreement	Frequency
1	11%	1	49%
2	18%	2	34%
3	44%	3	9%
4	15%	4	9%
5	4%	5	0%

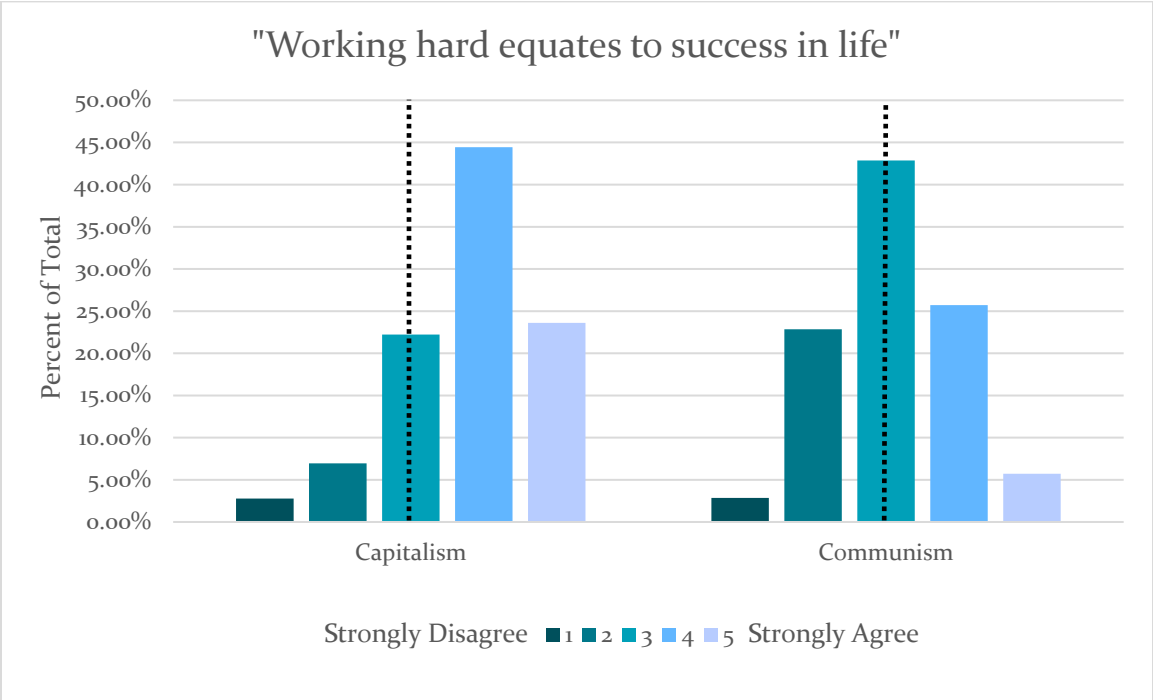
We can get a better view of general opinion by condensing 1 and 2 into "disagree", keep 3 as "impartial" and condensing 4 and 5 into "agree. After, we are left with the following chart.

Capitalist Agreement	Frequency	Communist Agreement	Frequency
Disagree	36%	Disagree	83%
Impartial	44%	Impartial	9%
Agree	19%	Agree	9%

From this chart we can gather some important facts. As a communist (from our sample at The New School), you are 2.3x as likely (130% more likely) to disagree with the statement compared to your capitalist counterpart. As a capitalist you are 5.2x as (420% more likely) likely to be ambivalent about the statement compared to your communist counterpart, and 2.3x as likely (130% more likely) to agree.

Does working hard equate to success?

There was a noticeable difference in the distribution of opinions when it came to the question of whether working hard equates to success in life. Capitalists more than twice as likely to believe that working hard equates to success in life, and communists were more than twice as likely to believe that working hard does not equate to success in life. Let's look at the data and get some exact figures.



When provided with a chart, it is much easier to see that while communist respondents are close to being normally distributed, capitalist respondents are quite negatively skewed. The descriptive statistics do not differ a huge amount between the capitalists and communists save for one or two exceptions, but I have included them here anyways.

Capitalist Stats	*Stats*	*Communist Stats*	*Stats*
Minimum	1	Minimum	1
Q1	3	Q1	2.5
Median	4	Median	3
Q3	4	Q3	4
Maximum	5	Maximum	5
Mean	3.79	Mean	3.09
Mode	4	Mode	3
Stand.Dev	0.98	Stand.Dev	0.92
Kurtosis	0.48	Kurtosis	-0.18
Skewness	-0.77	Skewness	0.06

While most of the statistics are similar, there is one statistic that I feel most accurately represents the difference properly. That is the skewness. While the skewness of the communist distribution fits with what would we would expect of a normal distribution, the capitalist skewness is quite negative. This suggests that it is more likely that capitalists will believe working hard equates to success in life. We can get a clearer picture of this by using a frequency table.

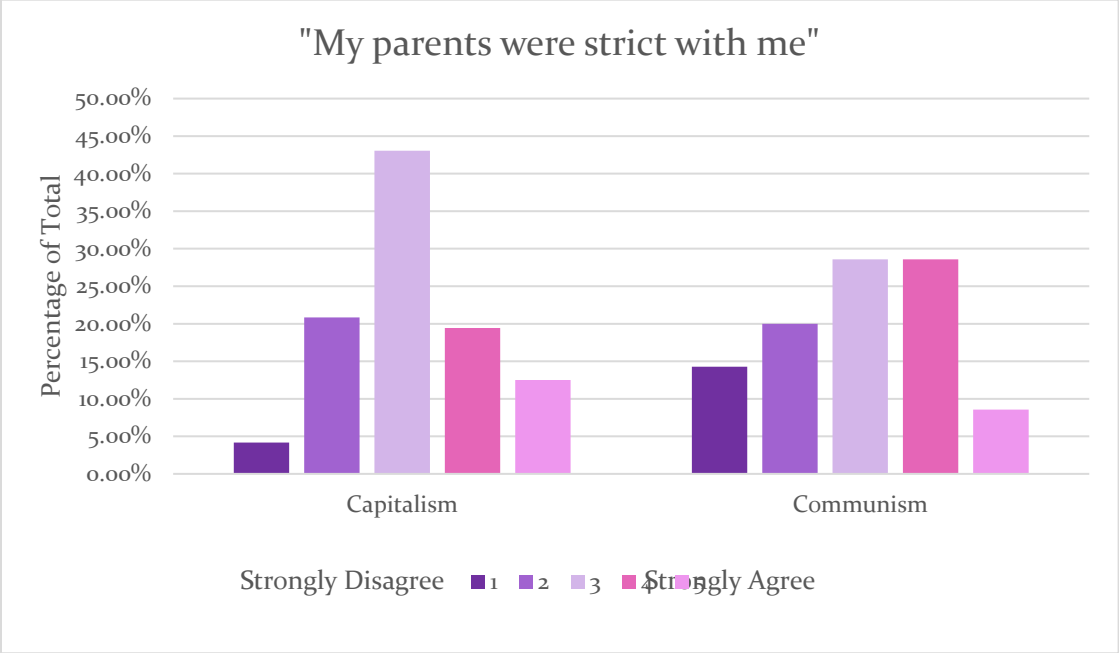
Capitalist Agreement	Frequency	Communist Agreement	Frequency
1	3%	1	3%
2	7%	2	23%
3	22%	3	43%
4	44%	4	26%
5	24%	5	6%

We can get a better view of general opinion by condensing 1 and 2 into "disagree", keep 3 as "impartial" and condensing 4 and 5 into "agree. After, we are left with the following chart.

Capitalist Agreement	Frequency	Communist Agreement	Frequency
Disagree	10%	Disagree	26%
Impartial	22%	Impartial	43%
Agree	68%	Agree	31%

From this simplification we can gather some important facts. As a capitalist (from our sample at The New School), you are 2.2x as likely (120% more likely) to believe that working hard leads to success in life compared to your equivalent communist counterpart. Meanwhile as a communist you are 2.6x as likely (160% more likely) to believe that working hard DOES NOT lead to success in life compared to your equivalent capitalist counterpart. As a communist, you are also 2x (100% more likely) as likely to be impartial about whether working hard leads to success in life than your capitalist counterpart.

Parental Strictness



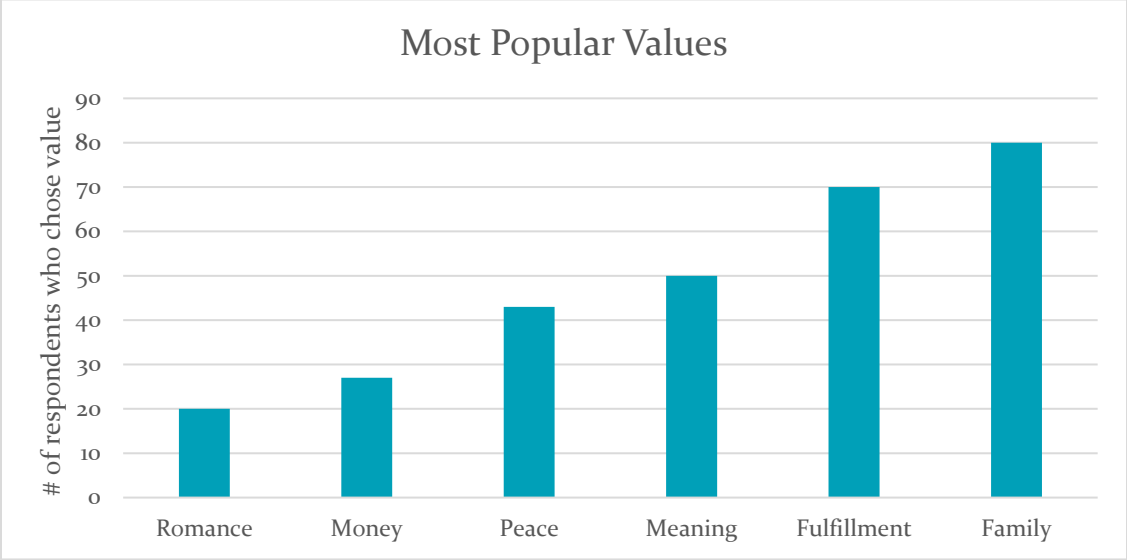
There was no significant difference between the distribution of capitalists and communists when it came to their perceived parental strictness. The capitalist distribution was slightly more negatively skewed, while the communist distribution was slightly more positively skewed. There was also a higher level of ambivalence in the capitalist distribution, however not to a particularly significant extent. We can guess that on average capitalists had stricter parents and communists had less strict parents, but the data isn't clear different enough between the two groups to confidently support this claim.

Values

To measure the difference in values between capitalists and communists the different values that people selected were separated into two groups: “capitalist values” and “communist values”. I then laid out all of the values that were chosen and, in each group, I counted the occurrence of those values. I then divided the count of the occurrence of those values by the total number of values chosen to adjust for the difference in the number of capitalist and communist respondents. We can than compare the percentage of communist respondents that chose certain values to the percentage of capitalist respondents for the same value.

There were certain values that needed to be omitted from the data, as there were not enough respondents who selected those values and therefore it would have presented an inaccurate picture of correlation. These are the values for which there were simply not enough respondents. Because of the proportional nature of the weights, if there was one respondent who chose a value on both sides, due to the high number of capitalist respondents, it would appear as though communists preferred that value twice as much. For instance, let us look at the value: “God” which was chosen once by both sides. On the communist side, god represents 1% of the values chosen by respondents, but on the capitalist side (due to twice the number of capitalist respondents) it is only 0.5% of the values chosen by respondents. As such anything with less than a total of 20 responses between both sides has been omitted. Note that after applying this filter, only 6 out of the original 23 value categories remained, however those 6 categories contained 91% of respondent data. The other categories did not have enough data to draw any meaningful conclusions from individually. These are however used at a later point to draw further conclusions from the data when grouped together.

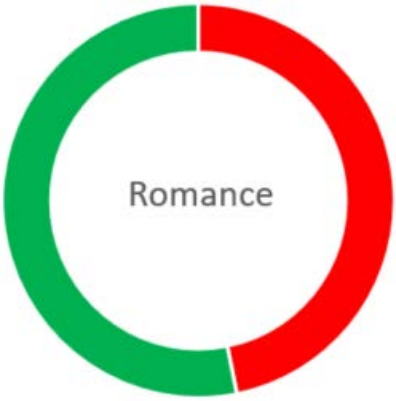
The six categories that remained after applying the filter were romance, money, peace, meaning, fulfillment, and family, and their popularity was distributed as follows:



Now by equally weighting capitalists and communists, we can see the percentage of people on either side that chose specific values, and determine the difference in preferred values between capitalists and communists.

The different weighted values have been displayed in pie charts below, where green is the percentage of capitalist respondents and red is the percentage of communist respondents.

As a capitalist student at The New School you are 10% more (1.1x as) likely to hold romance in your top three values than your communist equivalent.

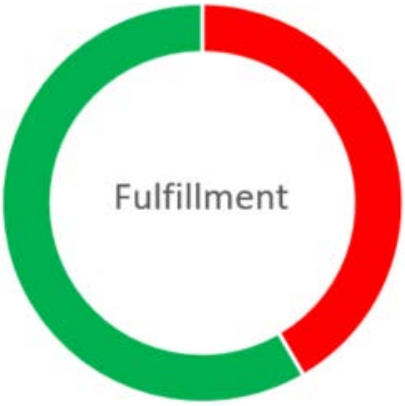


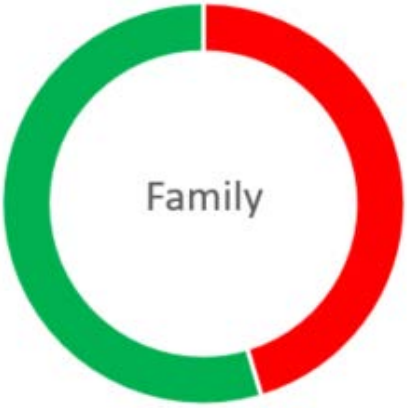
As a capitalist student at The New School you are 110% more (2.1x as) likely to hold money in your top three values than your communist counterpart.

As a communist student at The New School you are 60% more (1.6x) as likely to hold peace in your top three values than your capitalist counterpart.

As a communist student at The New School you are 40% more (1.4x) as likely to hold meaning in your top three values than your capitalist counterpart.

As a capitalist student at The New School you are 40% more (1.4x as) likely to hold fulfillment in your top three values than your communist counterpart.





As a capitalist student at The New School you are 20% more (1.2x as) likely to hold family in your top three values than your communist counterpart.

If we take the values we excluded earlier and include them in the data again, there is another analysis of the values data we can perform. We can clump the values that were given into the "given values" category, and we can clump people who added their own values using the other option into an "other" category.

Generally, people who are more conformist will be more accepting of what is given to them, while people who are non-conformist will try to find alternate options. Let's assume that the respondents who chose to select "other" were not just dissatisfied with the options given but saw the opportunity to add their own values and so in a nonconformist fashion took that opportunity. Along that line of logic, the respondents who preferred to just choose from the values that were given to them could be seen as more conformist. If so, we would expect the communists to have selected "other" much more than the capitalists seeing as it is more conformist to be capitalist in New York City than to be communist. As it turns out, this is the result. (Note that this value measurement, unlike the others, should be taken with a grain of salt as it is based on the assumption that nonconformists will choose "other" more often than conformists.)

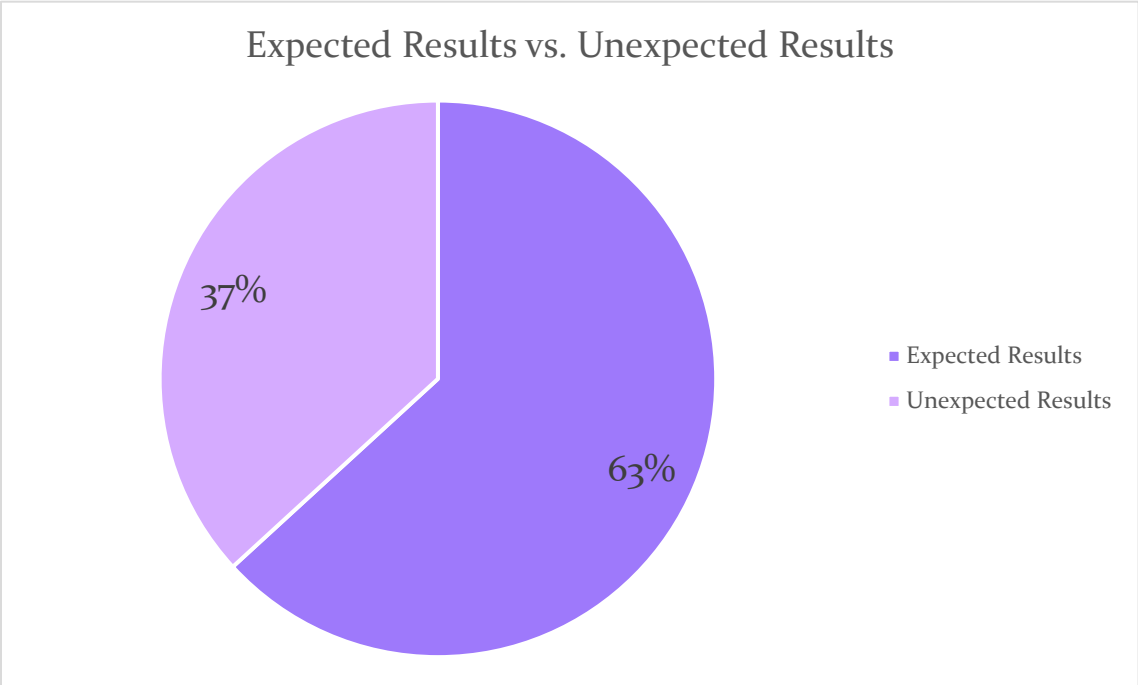
As it turns out, you are 190% more (2.9x as) likely to select "other" as a communist than a capitalist. Does this mean communists are non-conformist? Not at all! This test was based on a made-up assumption which only "seems" logical to me. Nonetheless, the fact that this is the largest disparity between capitalists and communists in terms of (potential) values, signifies there is certainly some interesting difference here.

Universal Equality vs. Individual Freedom

In a book I was recently reading (the title escapes me, I've been reading a lot recently and have lost track of which book I pick up ideas from), a hypothesis was posed that rather than the right and the left, or capitalism and communism, there are instead two warring value systems, and people who organize themselves into groups around either capitalism and communism or even right and left are instead organizing themselves by these warring value system. This "warring value system" is just a dichotomy of values, and that is whether the individual believes more in individual freedom, or universal equality. Individual freedom would equate to capitalism and the right in this case, and universal equality would equate to communism and the left.

To visualize the results, we need to categorize the data. If we weight the number of respondents who chose capitalism and communism equally, and we visualize the count of expected values (1) vs the count of unexpected values (0), we get this result.

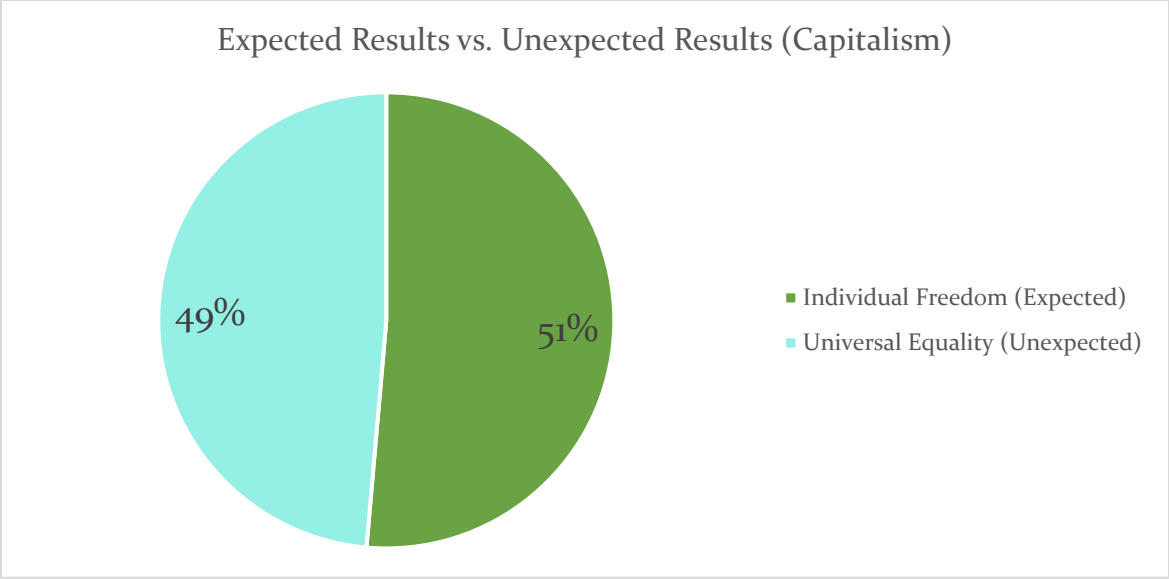
	Expected Results	**Unexpected Results**
Capitalism	Individual freedom (1)	Universal equality (0)
Communism	Universal equality (1)	Prefers individual (0)



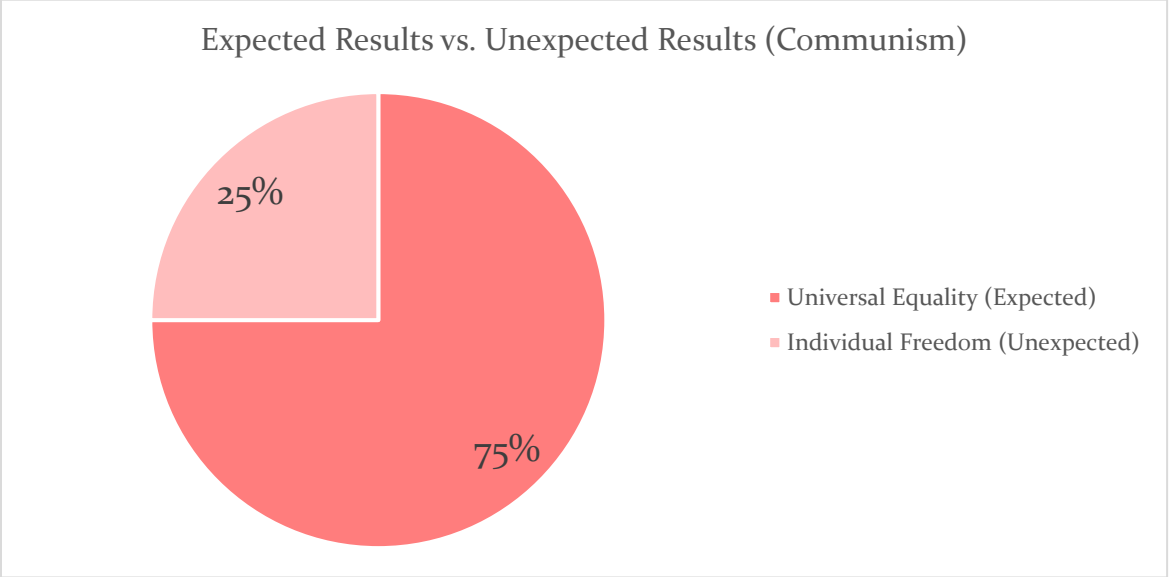
In this case the expected result is that capitalists believe in individual freedom more than universal equality AND communists believe in universal equality more than individual freedom. That result made up 63% of our data. The unexpected result is the exact opposite of the expected result.

If there was no correlation at all we would expect the pie chart to be split down the middle 50/50. If there was a strong correlation that confirmed our hypothesis, we would expect close to 100% of the graph to be our expected results. If there was a strong correlation that went exactly against our hypothesis (communists prefer individual freedom, capitalists prefer universal equality), then we would expect close to 100% of the graph to be our unexpected results. As it is, there appears to only be a very slight correlation, nothing that couldn't be explained by statistical anomaly. Certainly at the moment we cannot be confident in our expected result. Perhaps by separating communism and capitalism we can expect to see a clearer picture however?

Capitalism



Communism



Looking at just the expected and unexpected results for capitalism, we can see that there is perfectly no correlation between the variables. A capitalist is just as likely to hold individual freedom as a higher value than universal equality compared to vice versa (at least from our sample at The New School).

In examining the correlation present with communists, we get a more interesting picture. For every 1 communist to pick individual freedom as their value we get 3 communists picking universal equality. As a communist at The New School, we can be quite confident that you are around 3x as likely (200% more likely) to value universal equality over individual freedom.

CONCLUSION

Potential Biases

Some examples of potential bias throughout the study include:

- Because I am a capitalist, I have capitalist values, and would've given capitalist values as options in the survey, this might have biased the values choice against communists.

- The question that seeks to determine the strictness of the respondent's parents doesn't actually determine the strictness of the respondent's parents, it discovers how strict the respondents perceived their parents to be. If the parent was extremely strict but taught good values and did a good job raising their child, the child might see their parents as actually being quite easy going. On the other hand, if the parent wasn't strict at all, but rather treated their child horribly, the respondent might've interpreted that as strictness.

Survey/Study Revision

If I could revise the survey, I would probably be much more impartial about the answers. For example, I wouldn't ask if the US has been an overwhelmingly positive force, but rather I would ask what the respondents opinion on US foreign policy is.

I would also get a larger sample size, I feel like I could've gotten more accurate results, especially when examining age, if I had a larger sample size.

Results Analysis

While there may be a difference between capitalists and communists when it comes to some surface level opinions, opinions that can be changed by experience, the values respondents chose as their most important values were predominantly the very similar. The three most important values the respondents chose were family, meaning, and fulfillment. The difference of importance of these values between capitalists and communists were the smallest differences between capitalists and communists in this entire study. They were also the most important. I think the most important thing to take away from this study is that for the most part, we're all very similar, and we're all searching for the same things in life. If we work together towards these common values and look past our more superficial differences, we will all get much farther and be much happier.